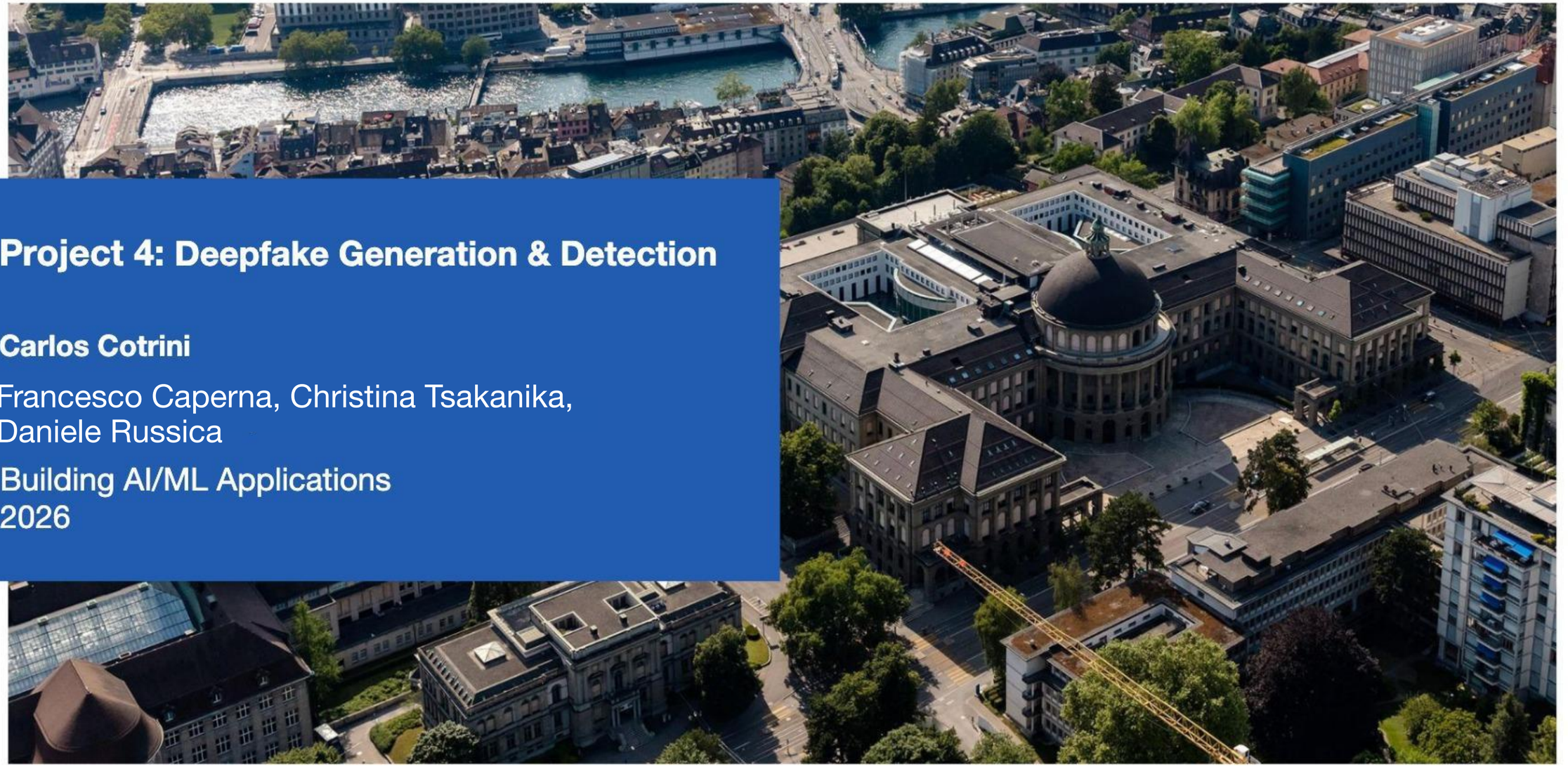
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Project 4: Deepfake Generation & Detection

Carlos Cotrini

Francesco Caperna, Christina Tsakanika,
Daniele Russica

Building AI/ML Applications
2026

Deepfake Generation & Detection

Image Inpaint Pipeline

Replace one fruit with another while preserving the background

Before



Original image with apple

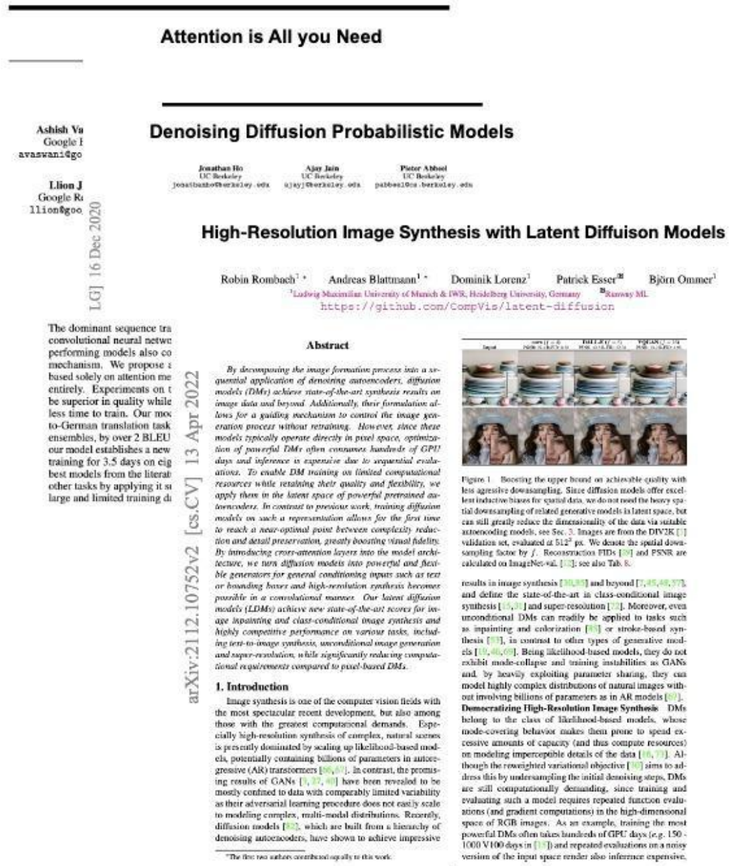
After



Generated image with orange

Reverse diffusion · T denoising steps · Conditioned on fruit embedding

Foundational Papers



Attention is All you Need

Denoising Diffusion Probabilistic Models

High-Resolution Image Synthesis with Latent Diffuison Models

Deepfake Generation & Detection

Goal

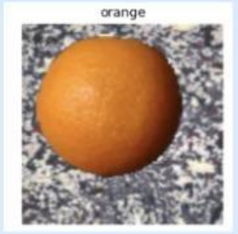
Train a U-Net to predict the noise pattern in corrupted images, conditioned on a fruit embedding
Run the reverse diffusion process to generate a new scene with the target fruit seamlessly in place

Dataset



4 versions per image

Composite	Mask	Redacted	Crop









The original scene as captured	Where is the fruit	Background scene context	The extracted fruit

Diffusion · Training

Phase 1 · Training

Input 1 Spatial context

Concatenated channel-wise before entering the U-Net

Noisy image + **Mask** + **Redacted**



x_t (B,4,16,16)

corrupted at random timestep t

(B,1,16,16)

where is the fruit?

(B,4,16,16)

background without fruit

cat (B, 9, 16, 16)

U-Net → **pred noise e_theta**

MSE loss = || ε - εθ ||^2

Input 2 Semantic conditioning

Which fruit should the model generate?



Fruit Encoder
CNN

→ **Embedding (B, 128)**

Crop (B,4,16,16)
the fruit alone

Cross-attention in U-Net

Each spatial position asks: 'what should I look like, given this fruit?'

Pipeline Overview

1 **Data Preparation**
composite, mask, redacted, crop

2 **VAE Encoder**
Pixel → latent space
composite, redacted

3 **Fruit Encoder**
crop → 128-dim embedding

4 **Diffusion U-Net**
Noise prediction
cross-attention

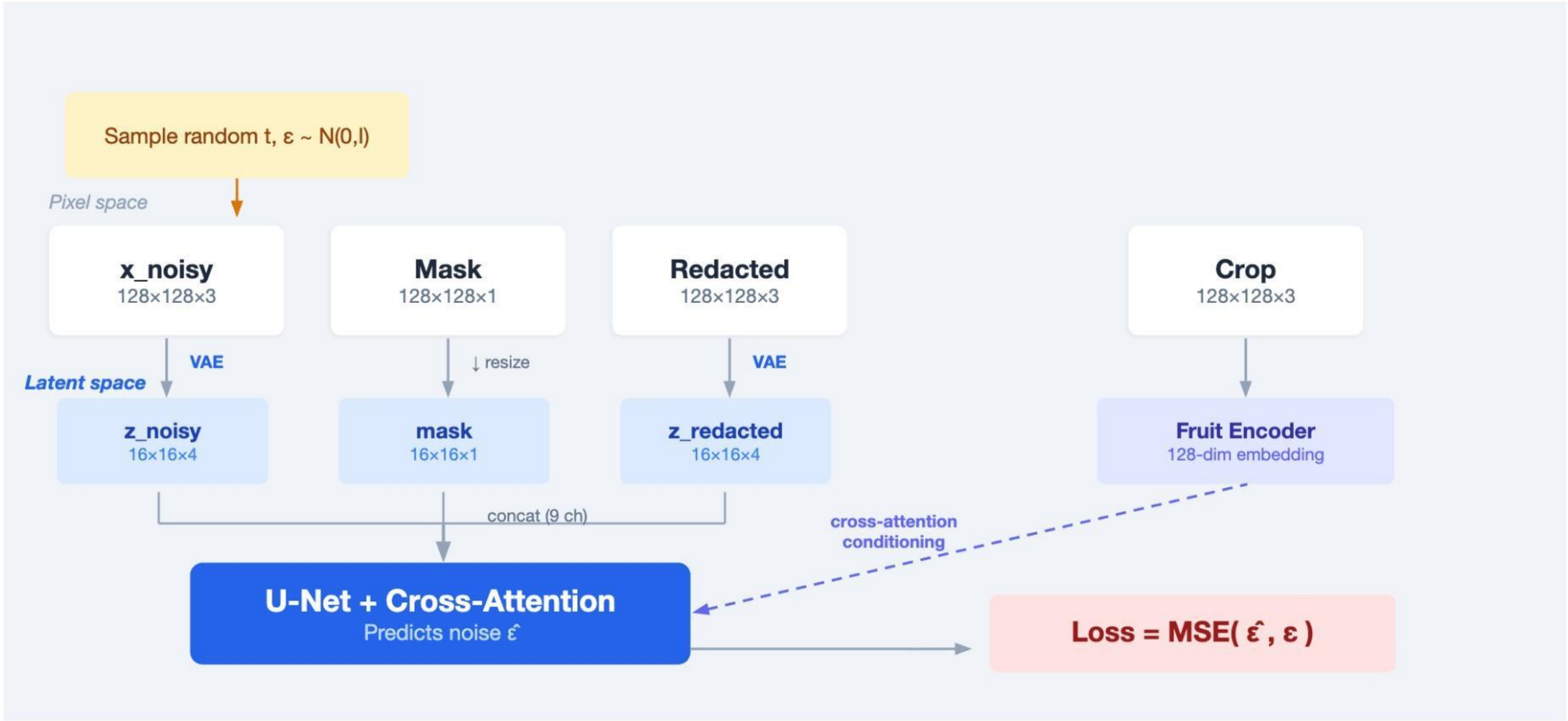
5 **VAE Decoder**
Latent → pixel space

6 **Discriminator**
Real vs fake classification

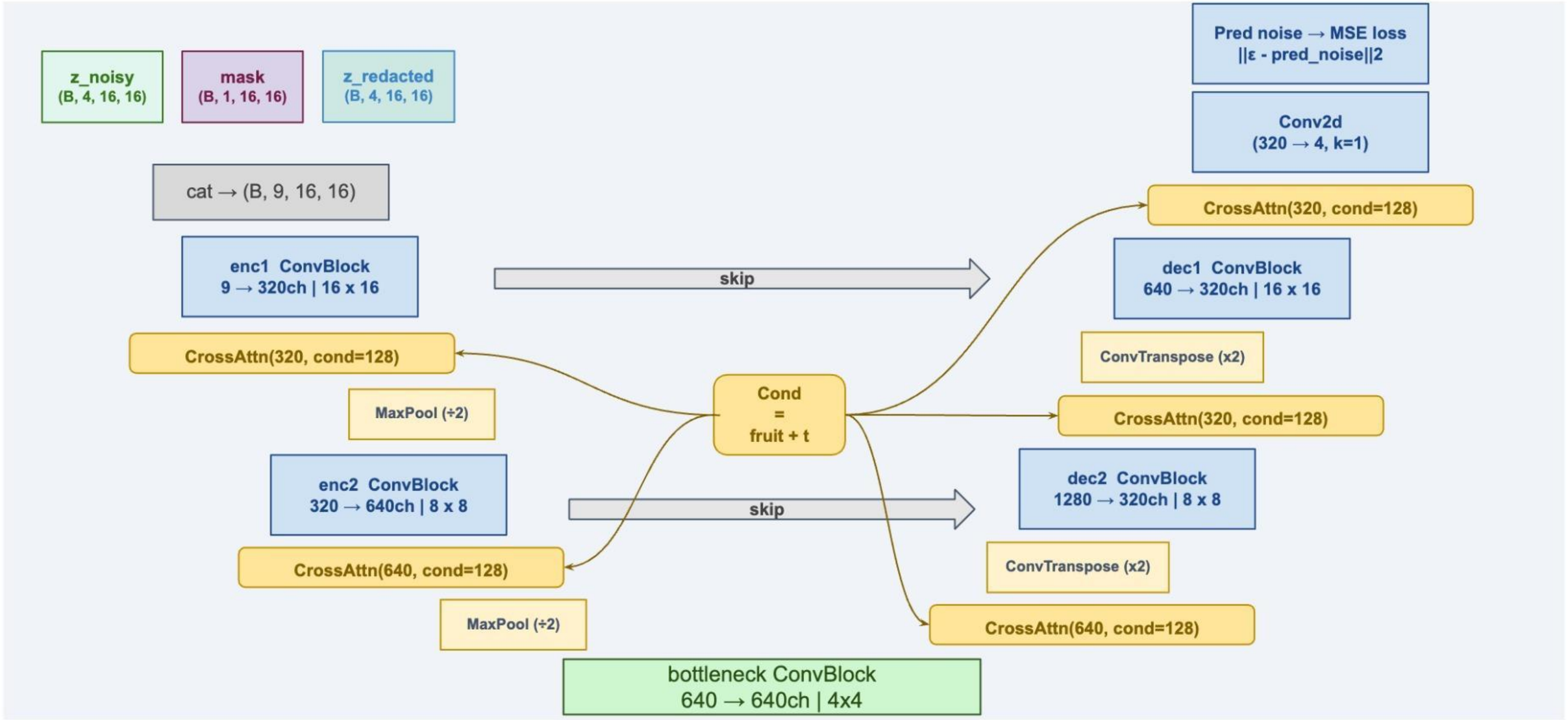
Generation (steps 1 to 5): a real scene becomes a deepfake. Encode to latent space, replace the fruit by reverse diffusion conditioned on the target embedding, decode back to pixels.

Detection (step 6) learns to catch them.

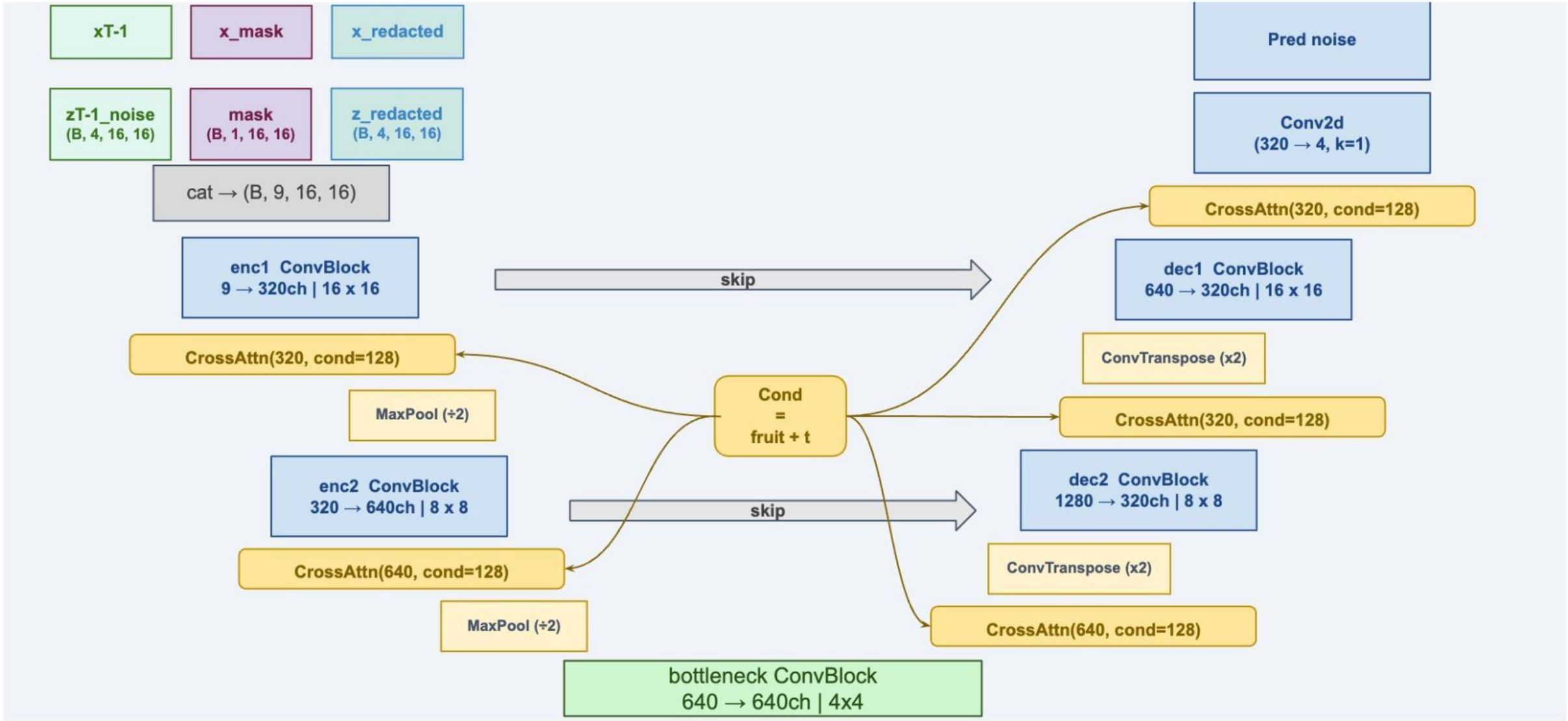
Training Process



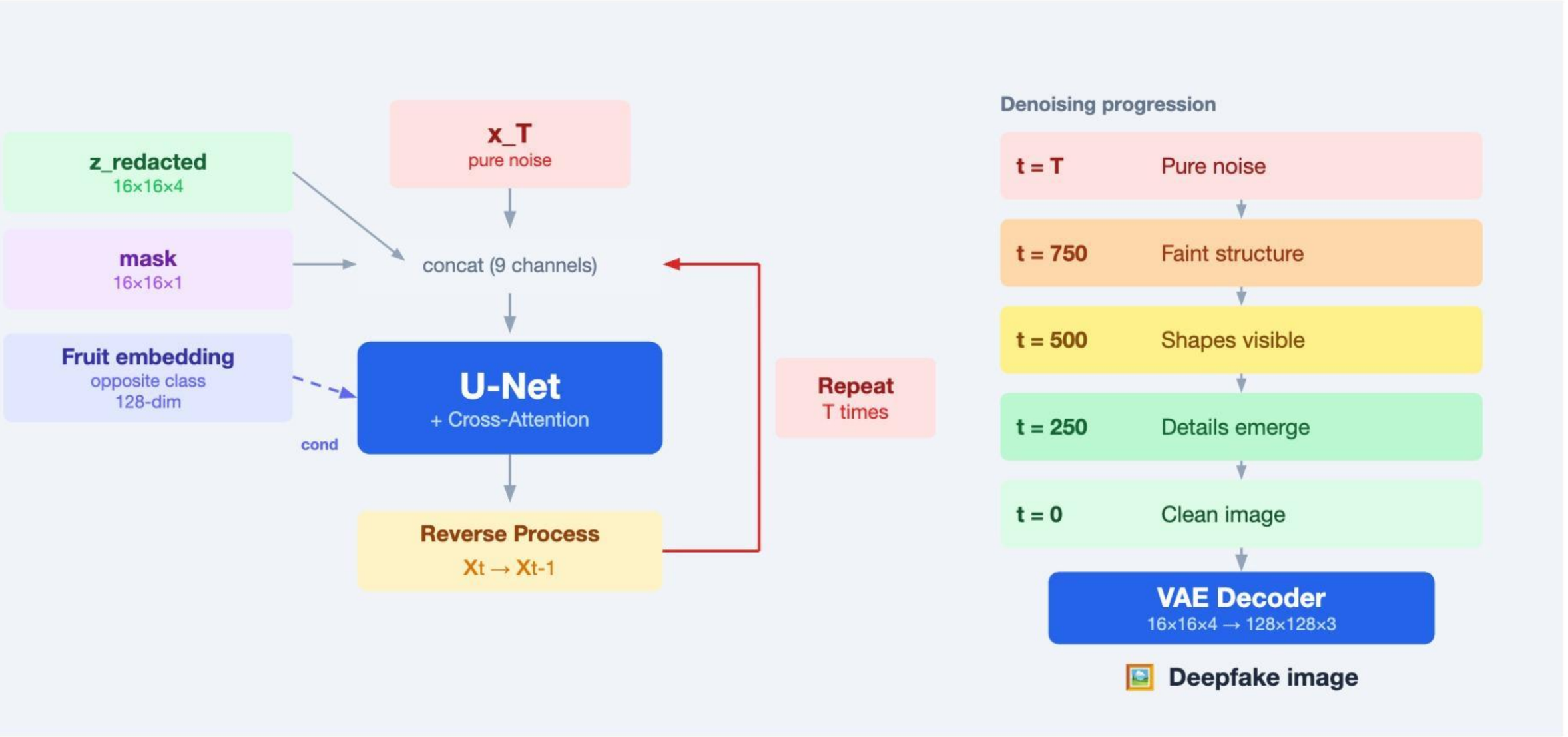
U-Net Training



U-Net Inference



U-Net Inference



Cross-Attention Mechanism

Feature map · Queries

Each latent position asks: “What should I look like?”

bg	bg	bg	bg
bg	**edge**	**edge**	bg
bg	**edge**	**center**	bg
bg	bg	bg	bg

Fruit embedding · Keys & Values

4 synthetic tokens projected from the 128-dim embedding

Token 1 Color	**Token 2** Shape
Token 3 Texture	**Token 4** Size

center	“What color and texture should I have?” → **strong** attention to the color and texture tokens
edge	“What contour should I follow?” → attention to the shape and size tokens
bg	**Weak** attention: fruit information is not needed here

Attention weight = softmax($Q \cdot K^T / \sqrt{d}$) · V. Line thickness in the original figure

Deepfake Detection

From forger to detective: can a model tell generated fruits from real ones?



STAGE 1 · LEARN WHAT REAL LOOKS LIKE

MODEL	Convolutional autoencoder
DATA	Real images only, no fakes
LOSS	MSE(reconstruction, image)
WHY	Fakes are scarce (100 vs thousands of reals). Fakes fall off the real manifold and reconstruct worse

STAGE 2 · CLASSIFY REAL VS FAKE

MODEL	Linear(128 → 1) on frozen embeddings
DATA	60 fakes + 60 reals, balanced
LOSS	Binary cross-entropy
WHY	Generators improve every month. Retrain only the tiny head when a new one appears: seconds, not hours

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Thank you

Questions?